K. J. Somaiya Institute of Technology

Department of Artificial Intelligence and Data Science

Project-Based Learning (PBL) – One Page Project Summary

Academic Year: 2026–27 | Semester: V

1. PROJECT DATA

Project Title:	Comparative Time-Series Forecasting of Food Inflation using El Niño Indices: ARIMA vs. Prophet vs. LSTM
Project Domain:	Machine Learning, Deep Learning, Time-Series Forecasting and Climate–Economic Analytics
Project Guide:	Prof. Vaishali Wadhe
Team Members:	Alihasan Zillani Vidhani (TY B-12); Ayush Surpal Tomar (TY B-8); Mudassir Siddiqui (TY B-2)

2. EXECUTIVE SUMMARY

This project evaluates whether the Oceanic Niño Index (ONI), an El Niño climate indicator, improves the forecasting of Indian food inflation. A 168-month dataset (January 2012–December 2025) is used to compare ARIMAX, Prophet, LSTM with ONI, and an inflation-only LSTM baseline. All four models are evaluated on January 2024–December 2025 data, and their predictions and error metrics are presented through a FastAPI-backed React dashboard.

3. PROBLEM STATEMENT

Historical inflation patterns may not capture climate-related influences on food supply and prices. However, adding ONI increases model complexity and may not improve accuracy. The project tests its incremental forecasting value against a historical-inflation-only baseline using a common evaluation period.

4. KEY OBJECTIVES

- Prepare a monthly Indian food-inflation series and align it with ONI data.
- Develop ARIMAX, Prophet, LSTM + ONI and inflation-only LSTM models.
- Compare 24 months of predictions using MAE, RMSE, sMAPE and MASE.
- Publish model results through REST APIs and a web-based dashboard.

5. METHODOLOGY & TECH STACK

Algorithms/Models:	ARIMAX (with lagged ONI), Prophet (ONI regressor), LSTM + ONI, LSTM Baseline.
Tools & Frameworks:	Python, Pandas, NumPy, Scikit-learn, Statsmodels, Prophet, TensorFlow/Keras, Jupyter, FastAPI, React, Vite and Recharts.
Dataset Used:	168 monthly observations from CPI/CFPI-derived food inflation and NOAA ONI (2012–2025); 24 test months in 2024–2025.
Hardware/ Environment:	Python/Jupyter development environment; web-browser dashboard and FastAPI service.

6. OUTCOMES & DELIVERABLES

Working Prototype/Model:	Four evaluated models, saved prediction files, a FastAPI backend and an interactive comparison dashboard.
Code Repository:	https://github.com/AyushTomarOG/Food-Inflation-in-El-Nino
Key Findings:	LSTM Baseline ranked first (MAE 1.3335; RMSE 1.6460), followed by LSTM + ONI (MAE 1.5245), ARIMAX and Prophet. ONI did not improve LSTM accuracy in this test; this does not rule out broader climate effects.

Project Guide Name & Signature: Prof. Vaishali Wadhe _____